

Using the Simulator

Overview

Cross Examination is a genetics mapping simulator with three tabs: MPP, Biparental QTL, and GWAS. Each tab simulates a different study design for mapping the genetic basis of a quantitative trait. The controls on the left set the experimental parameters; the plots on the right show the output after you click Simulate.

If you turn **Teaching Mode** ON before running, the true QTL positions and runs become fixed, so every student in the class sees the same result.

The MPP Tab — Controls

The MPP tab simulates a multiparent population (MPP) mapped with bulk segregant analysis (BSA). Cases are the top N individuals by phenotype; controls are a random N drawn from the non-extreme individuals. The CMH test combines signal across replicates.

Cross Examination | **MPP** | Biparental QTL | GWAS

☒ Teaching Mode (fixed QTL positions)

Population Design
Crossing design
Fully intercrossed

Founders
2 8 30

Generations of recombination/selfing
1 10 50

Pool
Cases = top N by phenotype. Controls = random N from unselected.
Individuals per pool (N)
200 2,000

Replicates (fresh phenotype draw per rep)
1 10

Each replicate re-draws phenotype noise and re-selects pools from the same RILs. CMH combines replicates: the QTL signal is consistent across reps (same genetics) so it stacks up, while non-QTL noise is random across reps and averages out. LOD at true QTL peaks rises and scan baseline variance falls as replicates increase.

Genetic Model
QTL architecture
Null - no QTL

Significance Threshold
Bonferroni (202 tests, alpha=0.05): LOD 3.8

Simulate

Save Plot

Design Tabs: MPP, Biparental QTL, & GWAS

Teaching Mode: check box for on position

Crossing Design: Fully Intercrossed or Hub-and-Spoke in the dropdown menu

Number of inbred Founders: slider to change values

Generations of recombination/selfing: slider to change values

Individuals per pool (N): the number of individuals sequenced in each pool of cases/controls

Number of Replicates: slider to change the number of independent replicates drawn

Genetic Model: Null – no QTL, 1 QTL, 2 QTLs, 3 QTLs, and Polygenic in the dropdown menu – same for all tabs

Simulate Button to simulate the design

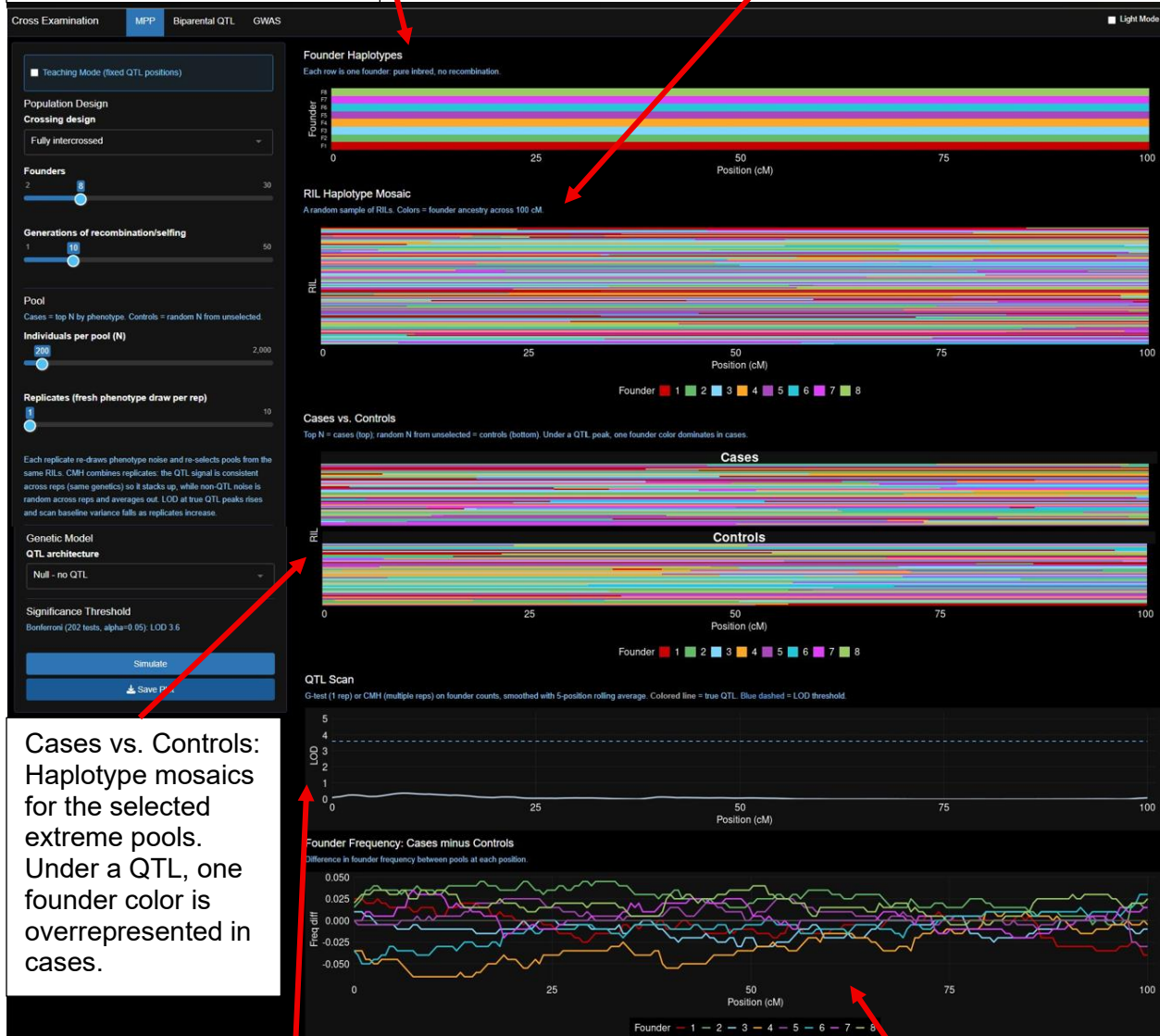
Save Plot Button to save the simulated design

The MPP Tab — Reading the Output

After you click Simulate, three panels appear on the right. The top two show population structure; the bottom two show the mapping results.

Founder Haplotypes: Each row is one founder line, fully inbred. Every position is fixed in that founder.

RIL Haplotype Mosaic: Each row is one RIL/individual (50 max plotted). The patchwork of founder colors shows where recombination changed ancestry.



Cases vs. Controls: Haplotype mosaics for the selected extreme pools. Under a QTL, one founder color is overrepresented in cases.

QTL Scan: LOD score at every genomic position using a G-test (or CMH test if multiple replicates). The colored vertical line marks the true QTL. The dashed line is the Bonferroni significance threshold.

Founder Frequency (Cases – Controls): For each founder at each position, the difference in haplotype frequency between cases and controls is plotted. Under a QTL, the founder carrying the effect allele shows a large positive shift.

The Biparental QTL Tab — Interface and Output

Two inbred parents are crossed to make F_1 , then intercrossed to make F_2 offspring. Each F_2 genome is a mosaic of the two parental haplotypes. A regression test at each position detects linkage between genotype and phenotype.

F_2 Individuals: the sample size for the QTL or the number of F_2 Individuals.

Parents panel: P1 (red) and P2 (blue) are fully inbred, and every locus is fixed.



F_2 haplotype mosaic: Each row is one F_2 individual. Alternating red/blue blocks show where recombination changed parental haplotypes.

QTL Scan: LOD score at each genomic position (smoothed). The colored vertical line marks the true QTL. The blue dashed line is the Bonferroni significance threshold.

The GWAS Tab — Interface and Output

Simulates a genome-wide association study (GWAS) in an outbred population. Individual genotypes are tested for differences in allele frequencies between cases and controls at each of 1,000 SNPs.

Individuals: the sample size for the GWAS or number per cases/controls.

LD block size: Controlled by the slider. Small blocks are more diverse, outbred populations, while large blocks are bottlenecked or recently hybridized species.

Light Mode: Makes the background white instead of black when enabled.



Cases vs. Controls: Each row is one individual. Colors show LD block haplotypes. Under a causal block, one color dominates in cases relative to controls.

Manhattan Plot: $\text{LOD} = \log_{10}(p)$ from a chi-square test at each SNP. All SNPs in a causal LD block rise together. The colored vertical line marks the true causal SNP; the blue dashed line is the Bonferroni threshold.